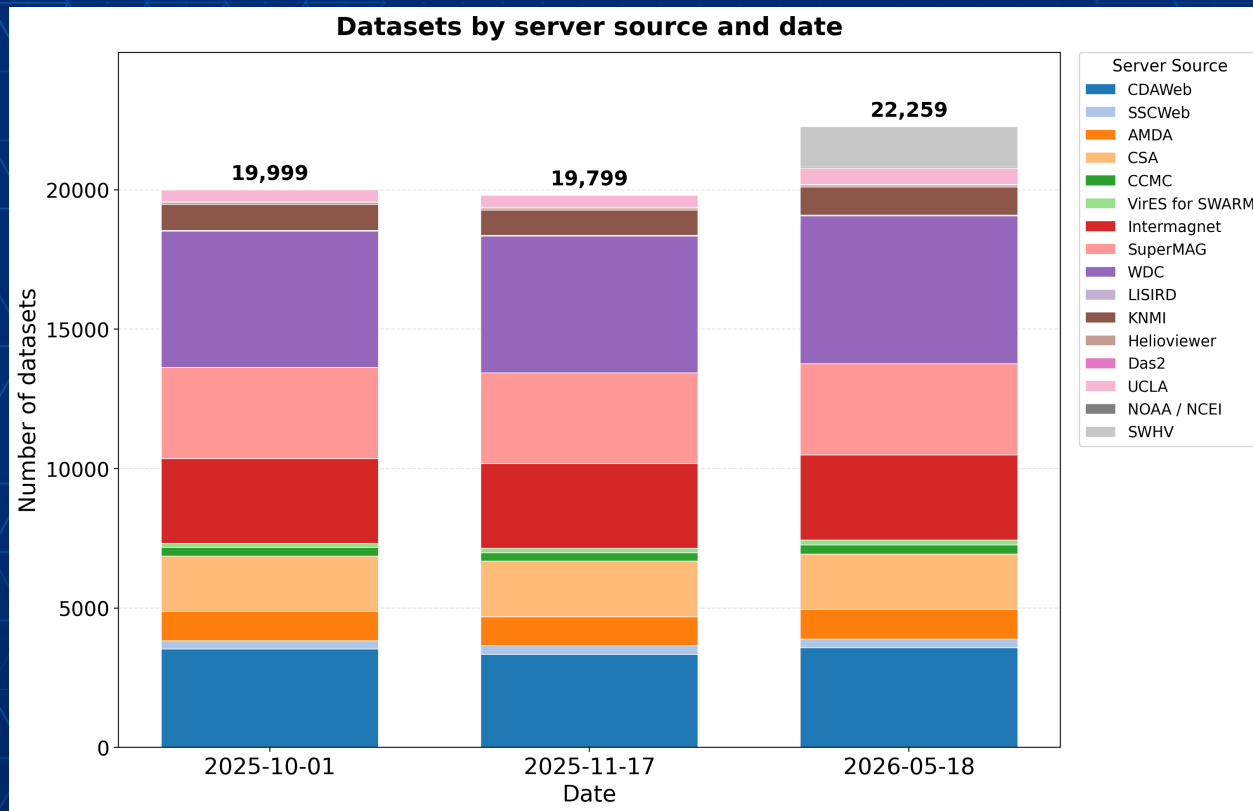


HAPI Status and Plans for PyHC, Spring 2026

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HAPI Server Adoption is increasing



Some servers are adding a few new datasets over time

New server: NOAA / NCEI

Others we hear about:
NOAA / SWPC (internal use)

Madrigal – very soon!
HamSCI (Personal Space
Weather Stations)– in progress
<https://pswsnetwork.eng.ua.edu/>

Detailed Dataset Counts as of 2026-05-19

Server	# Datasets		
	10/1/25	11/17/25	5/18/26
CDAWeb	3532	3344	3575
SSCWeb	301	301	313
AMDA	1044	1044	1065
CSA	1989	1989	1989
CCMC	300	300	317
VirES for SWARM	162	162	174
Intermagnet	3025	3033	3053
SuperMAG	3270	3270	3270
WDC	4889	4889	5309
LISIRD	29	29	29
KNMI	934	911	1003
Helioviewer	79	79	79
Das2 / Ulowa	19	19	19
UCLA	426	429	571
NOAA / NCEI			47
SWHV			1446
Total	19999	19799	22259

still verifying some aspects of these last two servers

NOAA NCEI Server

`https://www.ncei.noaa.gov/cloud-access/space-weather-portal/api/v1/hapi`

NCEI Space Weather Portal API

/cloud-access/space-weather-portal/api/v1/static/openapi/openapi-spot.yaml

API for accessing information and rich metadata for NCEI Space Weather data

Notes

- Information on each satellite and their products can be found on the associated [Web Portal](#).
- All requests need the major version prefix (eg "v1"). Major, non-compatible changes to the API may occur that will increment the major version number. Minor changes, fixes, and enhancements to the API will increment the minor build version only and not require a URL update.
- See example python plotting code using this API at the associated [Web Portal](#) help page.
- This server supports the **HAPI v3.2.0** specification for delivery of time series data. See the /hapi routes below. For more information, see the [HAPI github site](#)

Public Domain

v1 (build v20260413) OAS 3.0

id	title
f1m_dscovr	DSCOVr Faraday Cup 1 minute averages
f3s_dscovr	DSCOVr Faraday Cup 3 second resolution
fc0_dscovr	DSCOVr Faraday Cup Level 0 full resolution
fc1_dscovr	DSCOVr Faraday Cup Level 1 full resolution
m1m_dscovr	DSCOVr Magnetometer 1 min averages
m1s_dscovr	DSCOVr Magnetometer 1 sec resolution
mg0_dscovr	DSCOVr Magnetometer Level 0 full resolution
mg1_dscovr	DSCOVr Magnetometer Level 1 full resolution
mgc_dscovr	DSCOVr Magnetometer calibration file
att_dscovr	DSCOVr attitude
pop_dscovr	DSCOVr predicted orbit
rt0_dscovr	DSCOVr real time raw telemetry
tmd_dscovr	DSCOVr telemetry database
vc0_dscovr	DSCOVr raw telemetry
vc1_dscovr	DSCOVr stored raw telemetry
CCOR1_0A	CCOR-1 Level 0A
CCOR1_0B	CCOR-1 Level 0B
CCOR1_1A	CCOR-1 Level 1A
CCOR1_1B	CCOR-1 Level 1B
CCOR1_2	CCOR-1 Level 2
CCOR1_3	CCOR-1 Level 3
... total of 47 ...	

Follows OpenAPI spec and HAPI 3.2.0; has many other features.

We are still checking this out for compliance.



HAPI Servers

- node.js server for “pass-through” capability
 - handles all HAPI features and calls your command line program to get numbers from your data system
 - <https://github.com/hapi-server/server-nodejs>
- Python server
 - still the best general purpose server, but it needs a lot of work
 - <https://github.com/hapi-server/server-python>

=> NEW <=

- Cloud-based server – reads Parquet files in AWS cloud
 - developed by Eelco Doornbos at KNMI
 - <https://gitlab.com/KNMI-OSS/spaceweather/parquet-hapi-server>

HAPI Clients

- web-based exploratory tool
 - <https://hapi-server.org/servers/>
- Python client –or programmatic access
 - <https://github.com/hapi-server/client-python>
- KNMI Timeline Viewer
 - <https://spaceweather.knmi.nl/viewer/>
 - with images: <https://spaceweather.knmi.nl/viewer/?layout=viirs>
 - code repo: <https://gitlab.com/KNMI-OSS/spaceweather/knmi-hapi-timeline-viewer>
- next generation KNMI Timeline Viewer
 - <https://gitlab.com/KNMI-OSS/spaceweather/timeline-viewer>

Areas where HAPI is Evolving

Evolution 1: Adding Search for HAPI datasets

- Driver: users often ask: how do I find specific HAPI data?
- One solution: register all
- HAPI-to-SPASE metadata conversion process – mapping nearing completion
 - Madrigal will use this!
 - many other datasets are HAPI-only, so this will increase SPASE records dramatically
 - **this will be an automated (nightly?) process, allowing SPASE records to stay up to date!!**
 - SPASE team needs workflow for absorbing records
 - will be visible in HelioData portal at GSFC
 - <https://helio.data.nasa.gov/>
- In development: improved search mechanism
 - baseline is: <https://hapi-server.org/search/>
 - **needs documenting and better GUI**

Adding semantics to HAPI

- HAPI is so far about syntax – it tells you how to get tables of numbers, ordered by time
 - standard request syntax, standard response syntax for numeric, tabular data
- But, there are reasons to go beyond syntax to semantics and linkages
 - KNMI server already links datasets by cadence – can automate client request for the right data density
 - **search capabilities will return long lists of datasets – what are these and how do they relate to other datasets?!!**
 - multiple ground-based magnetometer datasets exist for the same station – how are they related?
 - alternate names - many datasets are supposedly the same but have different names
 - the overarching interest and need to find related content
 - provenance – what is the raw source of this calibrated data?
- We are adding a few features that address WHAT the data is, not just how to arrange it
 1. linkages between datasets
 2. schemas for a dataset

Note: Will be looking to IHDEA Semantics working group for standard terms!

Evolution 2: Linkages between HAPI datasets

- new endpoint to express connections or linkages between datasets

`server.org/hapi/relations` (*prototype, not 100% decided yet*)

- Examples:

datasetA **isAlternateCadenceOf** DatasetB

note: must consult each dataset to understand cadence differences

datasetC **isACopyOf** datasetD

- We will:
 - a. express these relationships using standard ontology language (turtle files)
 - b. provide an API for finding related datasets at a HPAI server

Evolution 3: HAPI Dataset Schemas

- a HAPI info response will contain an keyword telling what this dataset is
 - plain HAPI tells you: “these numbers have these labels”
 - HAPI with a schema tells you: “this table offers a standardized representation of a specific type of data”
 - examples: **File Listing**, Event List, Good-time intervals, Magnetometer data, etc
- the schema adds requirements to what the HAPI response contains
 - example: File Listing (red is required; blue is recommended; black is optional)

- **startDate**
- **uri**
- **modificationDate**
- **size**
- **checksum**
- **creationDate**
- **accessDate**
- **stopDate**

If your HAPI response contains these columns, it **isA** File Listing.

Column naming:
columns must have these names!
(we are considering allowing other names,
and then you have to give a mapping)

HAPI and agentic coding with AI

- because HAPI has a well-written spec, and lots of public examples, most AI models already understand it
- we have one-off stories of students using AI to code HAPI access routines
- need to make an MCP server for HAPI coding!
- low hanging fruit: a custom GPT for HAPI development
- open to suggestions for collaboration / collusion with PyHC chat capability!

HAPI Outreach and Activity

- Reaching out to new server groups: Madrigal, HamSCI, NOAA/NCEI, Belgian group (SWHV)
- GEM 2026, July12-17
 - focused, large presence of software and data providers negotiated with GEM steering committee
 - not just HAPI: SPEDAS / PySPEDAS, **PyHC**, HDRL, CCMC, SPDF, HelioCloud, Autoplot, Madrigal
 - presentations and tables at student day (Sunday)
 - lunch-time tables on Mon and Tue
 - if this works well, expand to other meetings (CEDAR, SHINE, AGU?)
- IHDEA Data Access Working group
 - had been HAPI-only, and therefore not really fully an IHDEA working group
 - being revitalized with new leadership: Jonathan C. (ESA), Jack I. (NASA), Jon V. (APL)
 - expect a survey soon looking for input and an invitation
- HAPI to SPASE converter
 - Madrigal will use this – they are standing up a HAPI server very soon!



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